

Object-Oriented Programming Using Java

Exception Handling

Contents

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References:

<https://docs.oracle.com/javase/tutorial/essential/exceptions/index.html>

What is an Exception?

- An *exception* is an event, which occurs during the execution of a program, that disrupts the normal flow of the program's instructions.

Example 1

```
14 public class ExceptionTest {  
15     public static void main(String[] args) {  
16         System.out.println(10/0);  
17         System.out.println("Hello!");  
18         System.out.println("Hello World!");  
19     }  
20 }  
21
```

Output - JavaApplication133 (run) X

run:
Exception in thread "main" java.lang.ArithmeticException: / by zero
at javaapplication133.ExceptionTest.main(ExceptionTest.java:16)
C:\Users\Nguyen Thi Thanh\AppData\Local\NetBeans\Cache\8.2\executor-snippets\run.xml:53: Java returned: 1
BUILD FAILED (total time: 0 seconds)

Example 2

```
3 public class ExceptionTest {
4     public static void main(String[] args) {
5         // Convert a string to integer
6         int n = Integer.parseInt("1.2");
7         System.out.println("n = "+n);
8         System.out.println("Ending Program.");
9     }
10 }
11
```

Output - ExceptionTest (run) ×

run:

Exception in thread "main" java.lang.NumberFormatException: For input string: "1.2"
at java.lang.NumberFormatException.forInputString([NumberFormatException.java:65](#))
at java.lang.Integer.parseInt([Integer.java:580](#))
at java.lang.Integer.parseInt([Integer.java:615](#))
at exceptiontest.ExceptionTest.main([ExceptionTest.java:6](#))
[C:\Users\Nguyen Thi Thanh\AppData\Local\NetBeans\Cache\8.2\executor-snippets\run.xml:53](#): Java returned: 1
BUILD FAILED (total time: 0 seconds)

Example 3

```
3 public class ExceptionTest {  
4     public static void main(String[] args) {  
5         int arr[] = new int[5];  
6         for (int i = 0; i <= 5; i++) {  
7             arr[i]=i;  
8         }  
9     }  
10 }  
11
```

Output - ExceptionTest (run) ×

run:

Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 5
at exceptiontest.ExceptionTest.main([ExceptionTest.java:7](#))

[C:\Users\Nguyen Thi Thanh\AppData\Local\NetBeans\Cache\8.2\executor-snippets\run.xml:53](#): Java returned: 1
BUILD FAILED (total time: 0 seconds)

Handling Exception by *try...catch...*

```
3 public class ExceptionTest {
4     public static void main(String[] args) {
5         int arr[] = new int[5];
6         try{
7             for (int i = 0; i <= 5; i++) {
8                 arr[i]=i;
9             }
10        }catch (Exception e){
11            System.out.println("Có lỗi vì: " + e);
12            System.out.println("Có lỗi vì: " + e.getMessage());
13        }
14    }
```

Output - ExceptionTest (run) ×

```
run:
Có lỗi vì: java.lang.ArrayIndexOutOfBoundsException: 5
Có lỗi vì: 5
BUILD SUCCESSFUL (total time: 0 seconds)
```

try block with more catch

```
4 public static void main(String[] args) {  
5     int arr[] = new int[5];  
6     try{  
7         arr[6]=100;  
8         int m = 10/0;  
9         int n = Integer.parseInt("1.2");  
10  
11     }catch (ArithmeticException e){  
12         System.out.println("Không thể chia một số cho 0");  
13  
14     }catch (NumberFormatException e){  
15         System.out.println("Dữ liệu chuyển đổi không hợp lệ");  
16  
17     }catch (ArrayIndexOutOfBoundsException e){  
18         System.out.println("Truy xuất chỉ số không hợp lệ");  
19     }  
20     System.out.println("Kết thúc chương trình.");  
21 }
```

Output - ExceptionTest (run) ×

run:
Truy xuất chỉ số không hợp lệ
Kết thúc chương trình.
BUILD SUCCESSFUL (total time: 0 seconds)

finally block

```
11 public class Main {  
12     public static void main(String[] args) {  
13         try{  
14             System.out.println(6/0);  
15         }  
16         finally{  
17             System.out.println("Finally");  
18         }  
19         System.out.println("End");  
20     }  
21 }  
22
```

The Finally block
always executes
when the try block
exists

Output - Test (run) x

```
run:  
Finally  
Exception in thread "main" java.lang.ArithmeticException: / by zero  
    at test.Main.main(Main.java:14)  
C:\Users\Nguyen Thi Thanh\AppData\Local\NetBeans\Cache\8.2\executor-snippets\run.xml:53: Java returned: 1  
BUILD FAILED (total time: 0 seconds)
```

How to *throw* an exception (1)

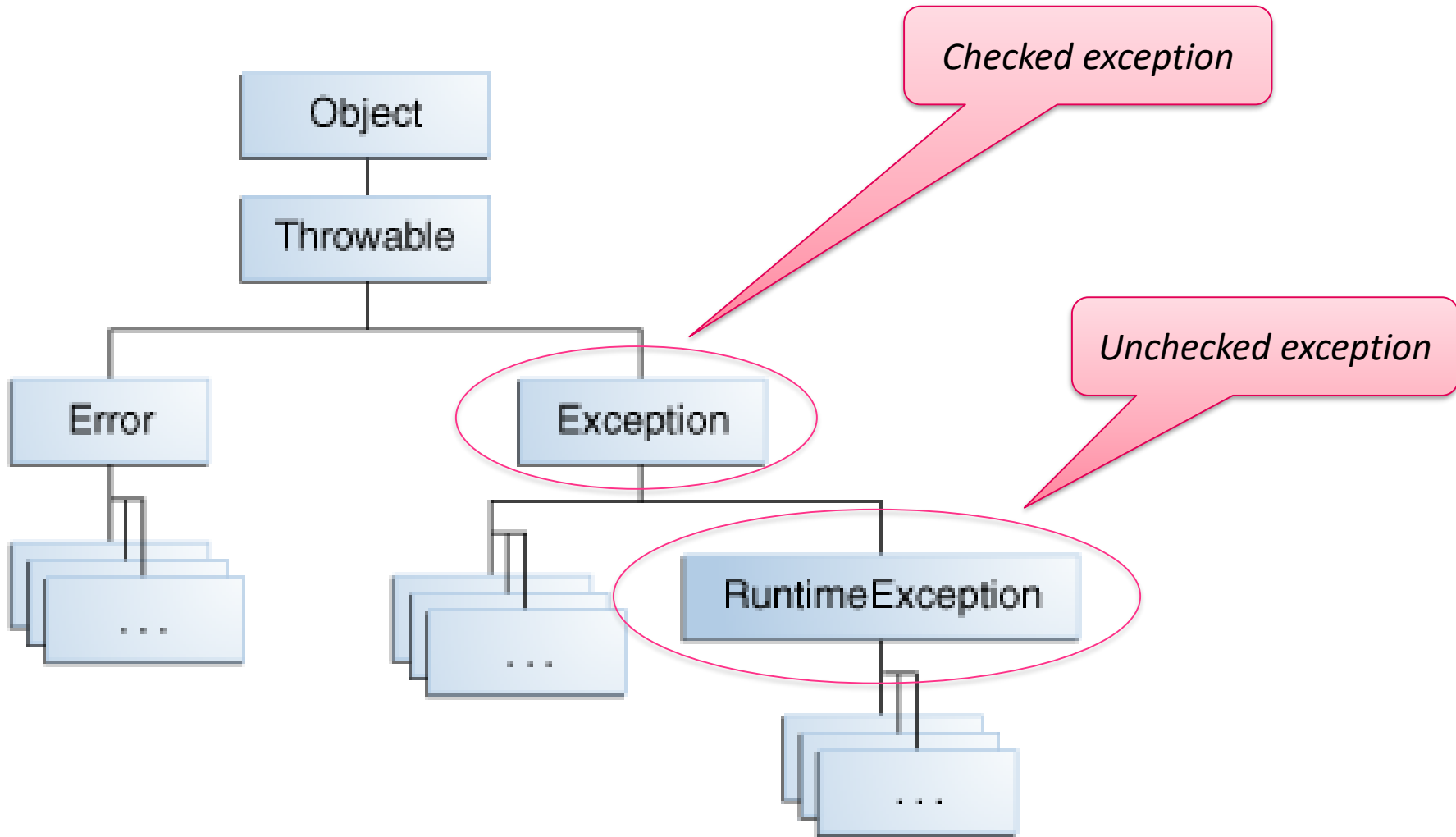
```
27  class Student{
28      String name;
29      int age;
30
31      public Student(String name, int age) throws Exception {
32          this.name = name;
33           setAge(age);
34      }
35      public void setAge(int age) throws Exception {
36          if (age<0) {
37              throw new Exception("Invalid age");
38          }else{
39              this.age = age;
40          }
41      }
42  }
```

How to *throw* an exception (2)

```
27 class Student{
28     String name;
29     int age;
30
31     public Student(String name, int age) throws Exception {
32         this.name = name;
33         setAge(age);
34     }
35     public void setAge(int age) throws Exception {
36         if (age<0) {
37             throw new Exception("Invalid age");
38         }else{
39             this.age = age;
40         }
41     }
42 }
```

```
public static void main(String[] args) {
    try{
        Student student = new Student("Tom", -18);
    }catch (Exception ex){
        System.out.println(ex.getMessage());
        System.out.println("Student can not be created");
    }
}
```

Two Kinds of Exception



Throw an *Exception*

```
12 public class Ex1 {
13     // loại Exception bắt buộc phải khai báo throws đầu hàm
14     public void m() throws Exception{
15         throw new Exception("Exception Message");
16     }
17     //bắt buộc dùng try...catch khi gọi X()
18     public void m2() {
19         try{
20             m();
21         } catch (Exception ex){
22             //code here
23         }
24     }
25     //hoặc tiếp tục khai báo throws đầu hàm
26     public void m3() throws Exception{
27         m();
28     }
29     // có lỗi
30     public void m1() {
31         m();
32     }
33 }
```

Throw an *RuntimeException*

- Không bắt buộc khai báo throw đầu hàm

```
12 public class Ex2 {  
13     public void m() {  
14         throw new RuntimeException("RuntimeException Message");  
15     }  
16  
17     // không bắt buộc try...catch hoặc throws đầu hàm  
18     public void m1() {  
19         m();  
20     }  
21 }
```

Example 4

```
27 class Student{
28     String name;
29     int age;
30
31     public Student(String name, int age) throws Exception {
32         this.name = name;
33          setAge(age);
34     }
35     public void setAge(int age) throws Exception {
36         if (age<0) {
37             throw new Exception("Invalid age");
38         }else{
39             this.age = age;
40         }
41     }
42 }
```

```
17 public class ExceptionTest {
18     public static void main(String[] args) {
19          unreported exception Exception; must be caught or declared to be thrown
20         ----
21         (Alt-Enter shows hints)
22
23         Student student = new Student("Tom", -18);
24     }
25 }
```

Output?

```
11 public class Main {  
12     public static void main(String[] args) {  
13         try{  
14             B();  
15             System.out.println("Hello");  
16             C();  
17         }  
18         catch(Exception e){  
19             System.out.println(e);  
20         }  
21     }  
22     static void B() throws Exception{  
23         throw new Exception("Exception");  
24     }  
25     static void C() {  
26         throw new RuntimeException("RuntimeException");  
27     }
```

Output - Test (run) X



run:



java.lang.Exception: Exception



BUILD SUCCESSFUL (total time: 0 seconds)

Output?

```
11 public class Main {  
12     public static void main(String[] args) {  
13         try{  
14             B();  
15             System.out.println("Hello"); //unreachable statement  
16         }  
17         catch(Exception e){  
18             System.out.println(e);  
19         }  
20     }  
21     static void B() throws Exception{  
22         C();  
23         System.out.println("Hello B");  
24         throw new Exception("Exception"); //unreachable statement  
25     }  
26     static void C() {  
27         throw new RuntimeException("RuntimeException");  
28         System.out.println("Hello C"); //unreachable statement  
29     }  
30 }  
31
```

Output - Test (run) X

```
run:  
java.lang.RuntimeException: RuntimeException  
BUILD SUCCESSFUL (total time: 0 seconds)  
>>
```

Output?

```
22 void A() {  
23     try{  
24         B();  
25     }  
26     catch(Exception e){  
27         System.out.println("Exception from A()");  
28     }  
29 }  
30 void B() {  
31     try{  
32         C();  
33     }  
34     catch(Exception e){  
35         System.out.println(e.getMessage());  
36     }  
37 }  
38 void C() {  
39     D();  
40 }  
41 void D() {  
42     E();  
43 }  
44 void E() {  
45     throw new RuntimeException("Exception from E()");  
46 }
```

run:
Exception from E()
BUILD SUCCESSFUL (total time: 10 seconds)

ExceptionTest t = new ExceptionTest();
t.A();

catch with no code

```
17 public class ExceptionTest {  
18     public static void main(String[] args) {  
19         try {  
20             //do something  
21         }  
22         catch (IndexOutOfBoundsException e) {  
23             //no code  
24         }  
25         catch (Exception e){  
26             System.err.println(e);  
27         }  
28     }  
29 }
```

Can be no code here